

Customer Shopping Behavior Analysis Report

Executive Summary

The purpose of this project was to analyze customer shopping behavior using transactional purchase data to identify trends in spending, product preferences, customer segmentation, and subscription behavior. The analysis leveraged Python for data preparation, MySQL for querying and business analysis, and Power BI for visualization.

The dataset contained 3,900 customer transactions across multiple product categories, enabling a comprehensive examination of purchasing patterns and customer demographics.

Objectives

The primary objectives were:

1. Understand customer purchasing behavior.
 2. Identify high-performing products and categories.
 3. Analyze subscription patterns.
 4. Segment customers based on purchasing history.
 5. Evaluate the impact of discounts on purchasing decisions.
 6. Create an interactive dashboard for decision-makers.
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Methodology

Data Preparation

The dataset was cleaned and prepared using Python.

Key preprocessing activities included:

- Handling missing values.
- Standardizing column names.
- Creating age-group classifications.
- Engineering purchase-frequency features.
- Removing redundant variables.

The cleaned dataset was then loaded into MySQL using mysql-connector-python.

SQL Analysis

Several business-focused SQL queries were developed.

Revenue Analysis

Revenue was analyzed across:

- Gender
- Age groups
- Subscription status

Customer Segmentation

Customers were categorized into:

- New Customers
- Returning Customers
- Loyal Customers

Product Performance

Analysis identified:

- Top-rated products
- Best-selling categories
- Products most dependent on discounts

Subscription Behavior

Subscription trends were examined to determine whether repeat customers were more likely to subscribe.

Dashboard Development

An interactive Power BI dashboard was developed to present insights visually.

Dashboard components included:

- KPI Cards
 - Category Sales Analysis
 - Subscription Distribution
 - Revenue by Age Group
 - Interactive Filters
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Key Findings

Customer Demographics

Young Adults generated the highest revenue contribution among all age groups.

Product Categories

Clothing emerged as the strongest-performing category in terms of sales and revenue.

Subscription Insights

The majority of customers were non-subscribers, highlighting an opportunity to improve subscription adoption.

Customer Loyalty

Customers with more purchase history demonstrated higher engagement and greater spending potential.

Discount Impact

Certain products showed a strong dependency on discounts, suggesting opportunities for pricing optimization.

Recommendations

Subscription Growth

Introduce subscription-exclusive rewards and personalized promotions.

Loyalty Programs

Implement tiered loyalty programs to encourage repeat purchases.

Product Promotion

Prioritize marketing efforts around top-rated and top-selling products.

Marketing Optimization

Target customer segments with the highest revenue contribution.

Pricing Strategy

Regularly assess discount effectiveness to maintain profitability while driving sales.

Conclusion

The project successfully transformed raw transactional data into actionable business insights through data cleaning, SQL analysis, and dashboard visualization. The findings can support strategic decisions related to customer retention, subscription growth, product positioning, and revenue optimization.